

RQL: A Portable Intermediate Representation for Retrieval Plans in RAG Systems

Technical report — proposal, prior-art grounding, and offline prototype
(arXiv-style draft; not a camera-ready venue submission)

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Draft status. Technical report / arXiv-style *proposal* + *prototype*. Claims are labelled [ESTABLISHED] (sourced prior art or verified docs), [HYPOTHESIS] (RQL design / packaging), [PROVISIONAL] (cited but not multimodally re-read), or [AUTHOR-ONLY] (paper tables we have not reproduced). Offline parser/planner/emit/E2E and a Colab synthetic FANNS smoke are feasibility evidence only. **No** fabricated experimental results; **not** claimed NeurIPS/VLDB acceptance or unreproduced ANN SOTA.

Abstract

Retrieval-augmented generation (RAG) systems compose embedding search, lexical retrieval, metadata and ACL filters, hybrid fusion, reranking, and sometimes graph or multi-hop expansion. In practice those stages are glued together in application code against vendor-specific vector-database APIs, so accuracy, latency, and reliability become properties of brittle scripts rather than of inspectable plans.

This technical report proposes **RQL (Retrieval Query Language)** as a **portable intermediate representation for retrieval plans**: a small algebra over scored evidence, cost-aware physical planning under recall, latency, and ACL constraints, and compilation through adapters to heterogeneous backends. The textual DSL is a convenient frontend; the centre of mass is the LogicalPlan/PhysicalPlan IR and the adapter doctrine [HYPOTHESIS].

Contributions. (1) A problem formulation for portable retrieval plans. (2) An algebra and physical operator set that *packages* established IR/DB mechanisms (RRF, Condorcet-fuse, linear/CC fusion, ColBERT MaxSim, PLAID/MUVERA late-interaction ladders, HyDE rewrite, MMR diversify, FANNS filter strategies, BlinkDB-style budgets) without claiming unreproduced author metrics. (3) Draft JSON Schemas (0.1.0-draft), a docs-only vendor API matrix, and an offline prototype stack (toy parser → physical planner → emit sketches → E2E CLI; 12/12 schema-validated toy runs). (4) Related-work positioning with multimodal re-reads of the key priors. (5) An honest evaluation split: what we measured vs what remains for citation-grade ANN/RAG.

Status. We report unit tests, schema validation, and offline emit sketches as *feasibility* evidence. A Colab synthetic FANNS PRE/POST smoke is plumbing only—**not** P0 and **not** citation-ready ANN quality. We do **not** report unreproduced large-set FANNS, BEIR, or RAG end-to-end scores. Claims throughout are labelled [ESTABLISHED], [HYPOTHESIS], [PROVISIONAL], or [AUTHOR-ONLY].

1 Introduction

1.1 Motivation

Modern RAG stacks must retrieve *evidence*—chunks, passages, entities, paths—under mixed modalities: dense vectors, sparse/BM25 channels, structured predicates (including ACLs), and increasingly late-interaction or graph expansion [Lewis et al., 2020] [ESTABLISHED]. Vector databases expose powerful but **non-portable** interfaces (JSON filters, GraphQL operators, SQL extensions, Redis commands, Cypher search). As a result, three operational properties leave the plan and hide in glue code:

Accuracy.

Hybrid fusion (e.g. reciprocal rank fusion [Cormack et al., 2009]), over-fetch+rerank, and multi-query rewriting improve recall only when stages are explicit and tunable [ESTABLISHED] as IR practice; RQL makes them plan nodes [HYPOTHESIS].

Latency. Filter placement (pre vs. post vs. subgraph vs. specialised label graphs vs. iterator ANN) and ANN parameters dominate end-to-end cost [Patel et al., 2024, Gollapudi et al., 2023, Zhang et al., 2023, Lin et al., 2025] [ESTABLISHED].

Reliability.

Silent empty results, client-side ACL filtering, and irreproducible efSearch folklore are operational failures; plans with EXPLAIN and mandatory predicate pushdown targets are the proposed remedy [HYPOTHESIS].

1.2 Idea in one paragraph

RQL is a portable IR for retrieval plans—an algebra over scored multisets of evidence together with a Cascades/Calcite-inspired physical planner that chooses filtered-ANN and fusion strategies under explicit recall/latency/ACL contracts, then compiles via polystore-like shims to concrete engines [Begoli et al., 2018, sub, 2024, Duggan et al., 2015] [HYPOTHESIS] (adjacency; not an embedding of those systems). The textual DSL is a frontend, not the contribution’s centre of mass.

1.3 Contributions

1. **Problem formulation** for portable, ACL-safe, budget-aware retrieval plans (Section 4).
2. **Hypothesized retrieval algebra and physical operator set** that packages established mechanisms (RRF, Condorcet, linear/CC, MaxSim, PLAID/MUVERA, HyDE, MMR, FANNS FilterExec modes, BlinkDB-inspired budgets) as one IR (Sections 5–6) [HYPOTHESIS]; prior mechanisms labelled [ESTABLISHED] / [AUTHOR-ONLY] where appropriate.
3. **Compilation doctrine + offline prototype:** docs-only vendor matrix (journal 0020), LogicalPlan/PhysicalPlan JSON Schema freeze 0.1.0-draft (journal 0021), toy parser / planner / emit / E2E CLI (journals 0022–0024, 0026) with 12/12 schema-validated toy runs (Section 7) [HYPOTHESIS].
4. **Related-work positioning** with multimodal re-reads of ACORN, VBASE, Filtered-DiskANN, FANNS survey, RRF/Condorcet/Bruch/Chen, ColBERT/PLAID/MUVERA, HyDE, MMR, BlinkDB, and Subtrait/Calcite adjacency (Section 3).
5. **Honest evaluation split:** unit/E2E/plumbing measurements vs remaining P0 (SIFT1M), P1 (live adapters), and judgment-bearing RAG work (Section 9)—no fabricated metrics.

1.4 Reader’s guide: how to implement / run RQL today

If you are an engineer who wants to *touch* the stack before reading the algebra:

1. **Skim the idea:** this section + Figure 3 (end-to-end offline pipeline) and Figure 2 (plan layers).
2. **Hello RQL (offline, no vector DB):**

```
cd experiments/harness
python rql_pipeline.py \
  --rql ../../examples/toy/01-hybrid-rrf.rql \
  --profile qdrant --validate \
  --out ../../experiments/results/e2e/my_run
```

Expect LogicalPlan JSON, PhysicalPlan JSON, and a vendor request *sketch* marked `notExecuted`.

Full toy matrix: four queries $\times$ three profiles = 12/12 (see `experiments/results/e2e/run_validate.txt`)

3. **Schemas:** `schemas/logical-plan.schema.json`, `schemas/physical-plan.schema.json` (0.1.0-draft).
4. **What is *not* ready:** live Qdrant/ES/pgvector smoke, SIFT1M FANNS P0, BEIR/RAG with judgments—see Section 9 and `experiments/HUMAN_TODO.md`.

More detail and file paths appear in Section 7.2.

1.5 Paper roadmap

Section 2 recalls ANN, hybrid fusion, late interaction, and RAG stages. Section 3 positions RQL against vector-DB APIs, FANNS, fusion, late interaction, IR languages, and portable-plan systems. Section 4 states requirements and non-goals. Sections 5–6 present the algebra and physical planning (plain language first, then formalism). Section 7 covers adapters, schemas, and the offline prototype. Section 8 maps common RAG patterns to example `.rql` files. Section 9 separates measured plumbing from citation-grade remaining work. Section 10 lists limitations; Section 11 gives actionable next steps. Appendix A is the reproducibility checklist.

2 Background

Short primers for full-stack and ML readers; IR/DB readers may skim. Formal RQL operators appear in Section 5.

2.1 Approximate nearest neighbour search

Graph indices such as HNSW [Malkov and Yashunin, 2018] and disk-oriented graphs (DiskANN line) underpin production vector search. Search quality/latency is controlled by parameters (e.g. candidate list size), yielding recall–throughput curves rather than exact SQL cardinalities [ESTABLISHED].

2.2 Hybrid vector + predicate search

Applications need ANN jointly with structured predicates. Common strategies include **pre-filtering**, **post-filtering**, specialised low-cardinality equality indices, and **predicate-subgraph** traversal as in ACORN [Patel et al., 2024] [ESTABLISHED] (see Section 3).

2.3 Hybrid lexical–dense fusion

Reciprocal Rank Fusion (RRF) combines ranked lists without score calibration [Cormack et al., 2009] [ESTABLISHED] (journal 0011). Condorcet-fuse sorts candidates by pairwise majority over input rankings [Montague and Aslam, 2002] [ESTABLISHED] as the majoritarian rank-only sibling (journal 0018); Cormack AUTHORS later prefer RRF over Condorcet on their suites [PROVISIONAL] as unreproduced comparison. Bruch et al. [Bruch et al., 2023] analyse the sibling *convex combination* of normalised lexical and semantic scores $f_{\text{Convex}} = \alpha \phi_{\text{Sem}}(f_{\text{Sem}}) + (1 - \alpha) \phi_{\text{Lex}}(f_{\text{Lex}})$ (especially TM2C2 with theoretical min–max ϕ_{tmm}) versus RRF, and argue RRF is parameter-sensitive while CC is sample-efficient to tune [ESTABLISHED] as literature mechanisms (journal 0016); author NDCG/Recall tables are unreproduced here [PROVISIONAL]. Chen et al. [Chen et al., 2022] prefer RRF over tuned min–max linear interpolation in a zero-shot hybrid setting (Recall@1K; journal 0017), disagreeing with Bruch’s TM2C2>RRF(60) NDCG claim — an [ESTABLISHED] setup-level literature conflict (see §6.6). Planner choice among Fuse_{rrf} , $\text{Fuse}_{\text{condorcet}}$, $\text{Fuse}_{\text{linear}}$, and Fuse_{ltr} remains workload-dependent [HYPOTHESIS].

2.4 Late interaction

ColBERT [Khattab and Zaharia, 2020] defines *late interaction*: independent encoders produce bags of contextualized embeddings E_q, E_d , scored by a sum of maximum similarities (MaxSim)

$$S_{q,d} = \sum_{i \in [|E_q|]} \max_{j \in [|E_d|]} E_{q_i} \cdot E_{d_j}^\top$$

(Eq. (3) in Khattab and Zaharia [2020]; cosine via L2-normalized dots; squared L2 also evaluated) [ESTABLISHED] (multimodal journal 0012). Document bags are offline-indexable; MaxSim is pruning-friendly for end-to-end retrieval. Subsequent engines add physical rewrites: PLAID’s centroid interaction and multi-stage centroid pruning funnel (candidate generation $\rightarrow$ pruned centroid MaxSim $\rightarrow$ unpruned centroid MaxSim $\rightarrow$ residual MaxSim) [Santhanam et al., 2022] [ESTABLISHED] as literature mechanisms (journal 0014; author latency tables unreproduced), and MUVERA Fixed Dimensional Encodings that reduce Chamfer/MaxSim search to single-vector MIPS plus MaxSim rerank [Dhulipala et al., 2024] [ESTABLISHED] as a literature mechanism (journal 0013); RQL compile packaging of the ColBERT $\rightarrow$ PLAID $\rightarrow$ MUVERA ladder remains [HYPOTHESIS]. Separately, HyDE [Gao et al., 2023] [ESTABLISHED] zero-shot dense retrieval by generating a hypothetical document with an instruction-following LM and retrieving neighbors of its Contriever embedding in document space (journal 0032); RQL REWRITE HYDE packaging remains [HYPOTHESIS]. Carbonell and Goldstein [Carbonell and Goldstein, 1998] [ESTABLISHED] Maximal Marginal Relevance (λ -relevance vs novelty) for diversity reranking and anti-redundant summarization (journal 0033); RQL DIVERSIFY MMR packaging remains [HYPOTHESIS].

2.5 RAG as a staged pipeline

A typical RAG retrieve path is: embed $\rightarrow$ search channels $\rightarrow$ filter $\rightarrow$ fuse $\rightarrow$ (rerank) $\rightarrow$ expand/traverse $\rightarrow$ generate. RQL aims to name these stages as plan operators rather than ad-hoc functions [HYPOTHESIS].

3 Related work

We separate [ESTABLISHED] mechanisms and survey structure from [HYPOTHESIS] RQL packaging and from [AUTHOR-ONLY] unreproduced empirics. Skim the subsection titles if you already know FANNS/RRF/ColBERT; the multimodal notes live under *journal/* and *knowledge/reads/*.

3.1 Vector database interfaces

Commercial and OSS systems expose vendor-specific query surfaces (metadata filters, hybrid search APIs, SQL extensions such as pgvector). A **docs-only adapter-surface survey** (2026-09-16; journal 0020; docs/09-vendor-api-matrix.md) tabulates filter DSLs, filter+ANN composition (PRE/POST/iterative/UNKNOWN), hybrid/BM25+dense, RRF and weighted fusion primitives, multi-vector/late interaction, query surfaces, and EXPLAIN/plan visibility for Qdrant, Elasticsearch/OpenSearch, Weaviate, Milvus, and pgvector (optional Pinecone/Redis) [ESTABLISHED] as a *documentation survey*—not a live bake-off. None of these vendors offers a cross-vendor ANSI-SQL analogue for multi-stage RAG plans [ESTABLISHED] (landscape notes + matrix). The hypothesis that RQL LogicalPlan compiles through negotiated shims onto these surfaces is stated in §7.4.

3.2 Filtered ANN / FANNS

Lin et al. survey (multimodal re-read). Lin et al. [Lin et al., 2025] formalise hybrid datasets/-queries and propose a pruning-focused taxonomy—**VSP**, **VJP**, **SJP**, **SSP**—classifying 17 algorithm

families (A1–A17; Figs. 1–2). They further argue that hybrid query difficulty depends on *selectivity and distribution* (ID/POD/OOD; Figs. 3–6), and highlight combining multiple FANNS algorithms with dynamic selection as a system-level direction (§6.3). Pass 1–5 notes (journal 0019; knowledge/reads/fanns-lin2025-2505.06501/) treat the **taxonomy and A1–A17 map** as [ESTABLISHED] survey structure grounding FilterExec packaging; author Fig. 3 recall curves remain unreproduced [PROVISIONAL].

ACORN (multimodal re-read). Patel et al. [Patel et al., 2024] argue that pre- and post-filtering are brittle under selectivity and query–predicate correlation, and that specialised equality indices constrain predicate cardinality. They propose denser, predicate-agnostic HNSW variants (ACORN- γ , ACORN-1) that *search the predicate subgraph* to approximate an oracle HNSW over surviving vectors. Our Pass 1–5 notes (journal 0006; OKF bundle under knowledge/reads/acorn-2403.04871/) treat author QPS claims as **not reproduced here** [ESTABLISHED] for the problem taxonomy; [PROVISIONAL] for unreproduced numeric speedups.

Filtered-DiskANN (multimodal re-read). Gollapudi et al. [Gollapudi et al., 2023] change *graph construction* (FilteredVamana / StitchedVamana) so edges respect geometry and labels, then search with label-aware greedy expansion. Pass 1–5 notes (journal 0010; knowledge/reads/filtered-diskann-www23/) ground FilterExec = SPECIALIZED for equality/label predicates; complex SQL-like filters remain an author-stated open problem [ESTABLISHED] for the specialised-graph idea; [PROVISIONAL] for unreproduced QPS curves.

VBASE (multimodal re-read). Zhang et al. [Zhang et al., 2023] formalise *relaxed monotonicity* and expose ANN traversal as Volcano Open/Next/Close inside PostgreSQL, enabling filter-during-traversal, multi-column TopK, range filters, and vector joins without a predicted over-fetch K' . Pass 1–5 notes (journal 0009; knowledge/reads/vbase-osdi23/) ground FilterExec = ITERATIVE and a concrete path for VSimJoin [ESTABLISHED] for the iterator architecture; [PROVISIONAL] for unreproduced latency claims.

3.3 Rank fusion (RRF and linear/CC)

Cormack et al. (multimodal re-read). Cormack, Clarke, and Büttcher [Cormack et al., 2009] define Reciprocal Rank Fusion as $\text{RRFscore}(d) = \sum_{r \in R} 1/(k + r(d))$ with $k = 60$ fixed after a pilot study. RRF is unsupervised and *rank-only* (no requirement that channel scores be calibrated), unlike CombMNZ. Pass 1–5 notes (journal 0011; knowledge/reads/rrf-cormack-sigir09/) treat the **formula and default k** as [ESTABLISHED] IR prior art for logical Fuse_{rrf}; author TREC/LETOR MAP tables are **not reproduced here** [PROVISIONAL] as unreproduced empirics.

Montague and Aslam [Montague and Aslam, 2002] introduce *Condorcet-fuse*: a majoritarian (Social Choice) fusion that sorts the candidate pool using a pairwise majority runoff over input rankings as the comparison function ($O(nk \log n)$; ranks only). Their Fig. 1 taxonomy places Condorcet-fuse beside Borda-fuse/rCombMNZ in the ranks-only / no-training cell, opposite CombMNZ on the score axis; weighted Condorcet and dependence filtering (correlated-run prune) are optional variants. Pass 1–5 notes (journal 0018; knowledge/reads/montague-aslam-cikm02-condorcet/) treat the **pairwise-majority sort mechanism and Fig. 1 fusion-input taxonomy** as [ESTABLISHED] literature for logical Fuse_{condorcet} as a Fuse-family sibling of Fuse_{rrf}; author TREC MAP / sign-test tables remain [PROVISIONAL] / AUTHOR-only. Cormack et al. later report RRF matching or beating Condorcet Fuse on their suites [ESTABLISHED] as an AUTHOR comparison (journal 0011) — motivating RRF as the portable default while retaining Condorcet as an optional majoritarian sibling [HYPOTHESIS].

Bruch et al. (multimodal re-read). Bruch, Gai, and Ingber [Bruch et al., 2023] analyse hybrid lexical–semantic fusion via a *convex combination* of normalised channel scores (TM2C2: theoretical min–max +

α) versus RRF (including parametric η_{Lex} , η_{Sem} and RRF-CC). They argue CC learning is largely agnostic among monotone normalisers, RRF is parameter-sensitive, and CC is sample-efficient; on their suite they report TM2C2 beating default RRF on NDCG (disagreement with Chen et al.). Pass 1–5 notes (journal 0016; knowledge/reads/bruch-arxiv-2210.11934/) treat the **CC/TM2C2 formulas, normalisation role, and RRF-vs-CC delineation** as [ESTABLISHED] literature mechanisms for logical $\text{Fuse}_{\text{linear}}$ (with Fuse_{ltr} as the learned multi-parameter end of the continuum); author Recall/NDCG tables remain [PROVISIONAL] / AUTHOR-only.

Chen et al. (cite-chase from Bruch [5]; multimodal re-read). Chen, Zhang, Lu, Bendersky, and Najork [Chen et al., 2022] study *zero-shot* hybrid retrieval and fuse lexical (+expansion) with deep (NPR) channels via RRF ($k=60$). They argue score linear interpolation needs min–max normalisation and α tuning that conflicts with zero-shot transfer; on Robust04 and TREC-COVID, their best-tuned min–max BM25+NPR interpolation still underperforms RRF(BM25, NPR) by about 3% relative Recall@1K (Fig. 2; primary metric Recall@1K/MAP, not NDCG). Pass 1–5 notes (journal 0017; knowledge/reads/chen-ecir2022-2201.10582/) treat the **RRF-for-zero-shot mechanism and the linear-interp critique** as [ESTABLISHED] literature; author Recall/MAP tables remain [PROVISIONAL] / AUTHOR-only.

Resolving Chen $\leftrightarrow$ Bruch (setup disagreement). The author-level disagreement is [ESTABLISHED] as a *literature conflict of setups*, not a single false claim: different metrics (Recall@1K vs NDCG), different linear forms (simple min–max α -interp vs TM2C2 ϕ_{tmm}), different objectives (zero-shot / no labels vs sample-efficient α with labels), and different channel/dataset suites. RQL packaging and the planner preference rule among $\{\text{Fuse}_{\text{rrf}}, \text{Fuse}_{\text{linear}}, \text{Fuse}_{\text{ltr}}\}$ remain [HYPOTHESIS] (see §6.6).

3.4 Late interaction / multi-vector retrieval

ColBERT (multimodal re-read). Khattab and Zaharia [Khattab and Zaharia, 2020] introduce contextualized late interaction: bags of BERT embeddings scored by MaxSim-sum (Eq. (3)). Pass 1–5 notes (journal 0012; knowledge/reads/colbert-sigir20/) treat the **MaxSim scoring definition and offline document-bag pattern** as [ESTABLISHED]; author MS MARCO MRR/latency tables are not reproduced here [PROVISIONAL].

PLAID (multimodal re-read). Santhanam et al. [Santhanam et al., 2022] accelerate ColBERTv2 late-interaction search with a four-stage funnel: centroid-based candidate generation, *centroid interaction* (approximate MaxSim over bags of centroid IDs using $S_{c,q} = CQ^\top$), *centroid pruning* by threshold t_{cs} , then residual decompression and exact MaxSim. Pass 1–5 notes (journal 0014; knowledge/reads/plaid-cikm22/) treat the **centroid-interaction / pruning pipeline (Fig. 5; Eqs. (2)–(5))** as [ESTABLISHED] literature mechanisms; author GPU/CPU speedup and MRR tables are not reproduced here [PROVISIONAL].

MUVERA (multimodal re-read). Dhulipala et al. [Dhulipala et al., 2024] reduce Chamfer/MaxSim multi-vector search to single-vector MIPS via Fixed Dimensional Encodings, then Chamfer rerank—contrasting PLAID’s multi-stage residual-preserving centroid pipeline [Santhanam et al., 2022]. Pass 1–5 notes (journal 0013; knowledge/reads/muvera-2405.19504/) treat the **FDE reduction + MIPS + MaxSim rerank mechanism** as [ESTABLISHED] literature; RQL rewrite packaging and unreproduced BEIR/PLAID latency numbers remain [HYPOTHESIS] / [PROVISIONAL].

HyDE (multimodal re-read). Gao et al. [Gao et al., 2023] factorize fully zero-shot dense retrieval *without relevance labels* as Hypothetical Document Embeddings: an instruction-following LM generates one or more hypothetical documents from query+instruction; an unsupervised contrastive encoder (Contriever/mContriever) embeds them; real documents are retrieved by MIPS in document space

(Eqs. (4)–(8); Fig. 1). Query–document similarity is not explicitly trained; the dense bottleneck is argued to filter hallucinations. Pass 1–5 notes (journal 0032; knowledge/reads/hyde-2212.10496/) treat the **hyp-doc** → **encode** → **doc–doc retrieve mechanism** as [ESTABLISHED]; author DL19/20, BEIR, and Mr.TyDi tables remain unreproduced [PROVISIONAL]. RQL’s REWRITE HYDE / Rewrite surface is [HYPOTHESIS] packaging of that pipeline (token-costed plan node feeding dense search), not a claim of reproduced IR gains.

MMR (multimodal re-read). Carbonell and Goldstein [Carbonell and Goldstein, 1998] define Maximal Marginal Relevance as greedy selection $\arg \max_{D_i \in R \setminus S} [\lambda \text{Sim}_1(D_i, Q) - (1 - \lambda) \max_{D_j \in S} \text{Sim}_2(D_i, D_j)]$ with $\lambda \in [0, 1]$ trading relevance ($\lambda=1$) against novelty/diversity ($\lambda=0$). Pass 1–5 notes (journal 0033; knowledge/reads/mmr-carbonell-sigir98/) treat the **MMR formula + λ extremes** as [ESTABLISHED]; author Table 1 sentence precision and SUMMAC F-scores remain unreproduced [PROVISIONAL]. RQL’s DIVERSIFY MMR / Diversify_{mmr} surface is [HYPOTHESIS] packaging of that formula (post-retrieve diversity node)—distinct from multi-channel fusion—and not a claim of reproduced summary/IR gains.

3.5 IR query languages

Classical IR languages (Indri, Galago, CQL/SRU) treat ranking as an operator algebra [met, 2004] [PROVISIONAL] (URL-verified references; deep re-read pending). RQL steals composability, not surface syntax fidelity [HYPOTHESIS].

3.6 Optimisers, portable plans, polystores

Subtrait [sub, 2024] [ESTABLISHED] a cross-language *serialized plan* IR (plans not SQL text), with extension relations/functions and only a *conventional* logical/physical split (docs pass: journal 0025; OKF knowledge/reads/subtrait-spec-2026-09/). Apache Calcite [Begoli et al., 2018] [ESTABLISHED] an embeddable algebra+optimizer over heterogeneous adapters, modeling physical properties as traits/calling conventions and citing Volcano/Cascades lineage [Graefe, 1995] (multimodal pass: journal 0025; OKF knowledge/reads/calcite-begoli-sigmod18/). RQL steals portable-plan and adapter doctrine as [HYPOTHESIS] adjacency—it does **not** claim to implement Subtrait or embed Calcite (§7.10). BigDAWG-style polystores [Duggan et al., 2015] motivate islands + shims [PROVISIONAL]. BlinkDB [Agarwal et al., 2013] [ESTABLISHED] the AQP idea of *dual declarative contracts*—either an error+confidence bound (ERROR WITHIN ϵ AT CONFIDENCE C) or a response-time bound (WITHIN T)—plus runtime Error-Latency Profile (ELP) sample-size selection over multi-resolution stratified samples (Pass 1–5: journal 0031; OKF knowledge/reads/blinkdb-eurosys13/). Author multi-TB latency/error tables remain unreproduced [PROVISIONAL]. RQL’s RECALL TARGET / LATENCY surface is [HYPOTHESIS] packaging inspired by that contract+ELP doctrine; it does **not** claim BlinkDB aggregate confidence intervals transfer to ANN recall@ k (§6.4).

3.7 Knowledge packaging (methodology, not runtime)

Google’s Open Knowledge Format (OKF) packages knowledge as markdown concepts with YAML frontmatter and graph-shaped links [okf, 2025]. We use OKF-*shaped* bundles for research notes. We do **not** claim OKF is part of the RQL execution runtime [ESTABLISHED].

3.8 What RQL is not

RQL is not “another vendor filter DSL”, not a guarantee of identical recall across HNSW libraries, and not a claim that SQL alone solves multi-stage RAG [HYPOTHESIS].

4 Problem formulation

4.1 Setting (plain language)

A RAG retrieve path needs a *plan*: which channels to search, how to apply mandatory security filters, how to fuse ranked lists, whether to rewrite or rerank, and which backend knobs to turn—all under latency and (approximate) recall budgets. Today that plan usually lives only in Python. We want one logical description that can compile to Qdrant, Elasticsearch/OpenSearch, pgvector, and kin, with explicit fallbacks when a backend lacks a feature [HYPOTHESIS].

4.2 Formal setting

Let collections store entities with payloads, optional graph links, and one or more retrieval channels (dense vectors, lexical fields, multi-vectors). A **retrieval plan** consumes a user/agent query (text and/or vectors), mandatory security predicates, and optional budgets, and returns a ranked evidence multiset.

4.3 Requirements

1. **Portability**: one logical plan maps to multiple backends via adapters.
2. **Filter correctness**: ACL/tenant predicates are hard constraints (fail closed).
3. **Explicit approximation**: recall/latency targets are first-class, not hidden knobs.
4. **Composability**: fusion, rewrite, expand, traverse, and similarity joins are plan nodes.
5. **Observability**: logical/physical EXPLAIN with provenance for evaluation harnesses.

4.4 Non-goals (this draft)

Identical numeric recall across ANN implementations; full DDL standardisation; replacing embedding model APIs; unreproduced SOTA claims; claiming NeurIPS/VLDB/arXiv acceptance on the strength of the offline prototype alone.

5 RQL algebra (original sketch)

All operators in this section are [HYPOTHESIS] unless a citation marks established prior art. Operator names and IR packaging are [HYPOTHESIS]; formulas and mechanisms credited to prior papers are [ESTABLISHED] or [AUTHOR-ONLY].

5.1 Plain language first

Think of a retrieval plan as a small dataflow:

1. **Leaves** search one channel (dense ANN, BM25, or late-interaction MaxSim) and emit scored hits.
2. **Filter** drops hits that fail metadata/ACL predicates (must never be stripped client-side for hard ACLs).
3. **Fuse** merges several ranked lists (RRF by default; optional Condorcet, linear/CC, or learned LTR).
4. **Optional stages** rewrite the query (e.g. HyDE), diversify, rerank, expand parents, or traverse a graph.

Channels keep their own scores until a fuse node—you do not add a cosine to a BM25 score without an explicit fuse policy. Figure 1 shows a minimal hybrid RRF program that the offline toy parser accepts.

```

-- Toy hybrid RRF (examples/toy/01-hybrid-rrf.rql)
RETRIEVE chunks
  SEARCH
    DENSE ON embedding METRIC cosine CANDIDATES 50
      QUERY 'portable retrieval IR for RAG'
      VECTOR_REF $q_dense
    AND BM25 ON content CANDIDATES 50
      QUERY 'portable retrieval IR for RAG'
  FUSE RRF K 60;

```

Figure 1: Minimal “hello RQL” listing accepted by the toy frontend (journal 0022). Hypothesis surface syntax; RRF formula with $k=60$ is [ESTABLISHED] [Cormack et al., 2009].

5.2 Typed evidence relation

An **evidence relation** E is a multiset of tuples with at least:

$$E \subseteq Id \times Payload \times Score \times Channel \\ \times Provenance \times Attr^*.$$

Informal signature: each tuple carries a channel-local score until a fuse operator. Mandatory ACL attributes (tenant, clearance, ...) are part of $Attr$ and must remain enforceable in every physical plan [HYPOTHESIS].

Channels. $Channel \in \{\text{dense, bm25, late, graph, fused, } \dots\}$. Scores are comparable *within* a channel; cross-channel comparison requires Fuse.

5.3 Logical operators (kernel)

Operator	Signature (sketch)	Notes
$\text{Search}_{dense}(q, k)$	$\rightarrow E$	ANN leaf
$\text{Search}_{bm25}(q, k)$	$\rightarrow E$	Lexical leaf
$\text{Search}_{late}(q, k)$	$\rightarrow E$	Late interaction / MaxSim-sum scoring [ESTABLISHED] [Khattab and Zaharia, 2020] (journal 0012); RQL op name / compile [HYPOTHESIS]
$\text{Filter}(P)$	$E \rightarrow E$	Predicates / ACL
Union	$E \times E \rightarrow E$	Multi-query
$\text{Fuse}_{rrf}(k_{rrf})$	$E^* \rightarrow E$	Formula $\sum 1/(k + r(d))$, default $k=60$ [ESTABLISHED] [Cormack et al., 2009] (journal 0011); RQL op name / compile [HYPOTHESIS]
$\text{Fuse}_{condorcet}$	$E^* \rightarrow E$	Pairwise-majority sort (Condorcet-fuse) [ESTABLISHED] as mechanism [Montague and Aslam, 2002] (journal 0018); RQL op / compile [HYPOTHESIS]
$\text{Fuse}_{linear}(\alpha)$	$E^* \rightarrow E$	Convex combo of normalised channel scores [ESTABLISHED] as mechanism [Bruch et al., 2023] (journal 0016); RQL op name / ϕ / compile [HYPOTHESIS]
$\text{Fuse}_{ltr}(\text{model})$	$E^* \rightarrow E$	Learned / multi-feature fuse beyond single α [HYPOTHESIS] (continuum with linear; not a Bruch TM2C2 claim)
$\text{Diversify}_{mmr}(\lambda)$	$E \rightarrow E$	MMR $\lambda \text{Sim}_1 - (1-\lambda) \max \text{Sim}_2$ [ESTABLISHED] [Carbonell and Goldstein, 1998] (journal 0033); RQL op / DIVERSIFY MMR / defaults [HYPOTHESIS]
Rerank_m	$E \rightarrow E$	Cross-encoder etc.
Expand	$E \rightarrow E$	Parent/window
Rewrite / Rewrite _{hyde}	$q \rightarrow q^*$ (or emb*)	HyDE hyp-doc $\rightarrow$ encode $\rightarrow$ doc-doc MIPS [ESTABLISHED] as mechanism [Gao et al., 2023] (journal 0032); RQL op / REWRITE HYDE surface / costs [HYPOTHESIS]; multi-query sibling still [PROVISIONAL]
Traverse_h	$E \rightarrow E$	Bounded graph hop
VSimJoin_θ	$E \times E \rightarrow E$	Similarity join

5.4 Rewrite examples (hypothesis, literature-grounded)

- **Post-filter brittleness (ACORN seed).** $\text{Filter}(P) \circ \text{Search}_{dense}(q, k)$ may be rewritten toward a physical SUBGRAPH or over-fetched POST plan when estimated selectivity / query–predicate correlation is adverse [Patel et al., 2024] [HYPOTHESIS] (choice is physical; logical Filter remains).
- **Iterator pushdown (VBASE seed).** When the adapter exposes Volcano-style ANN iteration, $\text{Filter}(P) \circ \text{Search}_{dense}$ may compile to $\text{FilterExec} = \text{ITERATIVE}$ (filter during Next) rather than TopK-then-drop [Zhang et al., 2023] [HYPOTHESIS].

- **Label-specialized graphs (Filtered-DiskANN seed).** Equality/label predicates with a capable filtered graph may compile to SPECIALIZED [Gollapudi et al., 2023] [HYPOTHESIS]; workload-stable partitions may compile to PARTITION and multi-algorithm backends to ROUTER [Lin et al., 2025] [HYPOTHESIS]; unsupported predicate shapes must not silently use specialised paths.
- **Hybrid fuse (RRF / linear / LTR).** Parallel Search_{dense} and Search_{bm25} (or other channels) remain separate until a fuse op. $\text{Fuse}_{rrf}(k_{rrf})$ with default $k_{rrf} = 60$ [Cormack et al., 2009] is the portable *rank-only* default when scores are incomparable [ESTABLISHED] (journal 0011). $\text{Fuse}_{condorcet}$ is the majoritarian rank-only sibling (pairwise-majority sort) [Montague and Aslam, 2002] [ESTABLISHED] as mechanism (journal 0018); offer when pairwise Condorcet semantics are desired, without displacing RRF as default [HYPOTHESIS] (Cormack AUTHORS prefer RRF on their suites). $\text{Fuse}_{linear}(\alpha)$ follows Bruch-style convex combination of normalised scores [Bruch et al., 2023] [ESTABLISHED] as mechanism (journal 0016; unreproduced NDCG [PROVISIONAL]); escalate to Fuse_{ltr} when a learned multi-feature model is justified [HYPOTHESIS]. Choosing among $\{rrf, condorcet, linear, ltr\}$ is physical/policy [HYPOTHESIS] (§6.6; Chen $\leftrightarrow$ Bruch setup disagreement [ESTABLISHED], journals 0016–0017).
- **Late-interaction leaf + rewrites (ColBERT / PLAID / MUVERA).** Logical Search_{late} carries MaxSim-sum semantics [Khattab and Zaharia, 2020] [ESTABLISHED]. Physical compile targets [HYPOTHESIS]: (i) native multi-vector MaxSim; (ii) PLAID-class centroid interaction + multi-stage prune then residual MaxSim [Santhanam et al., 2022] [ESTABLISHED] as the literature pipeline (journal 0014; unreproduced speedups remain [PROVISIONAL]); (iii) MUVERA FDE $\rightarrow$ MIPS then MaxSim/Chamfer rerank [Dhulipala et al., 2024] when only single-vector ANN is available (journal 0013). Missing multi-vector, PLAID, and FDE capabilities $\rightarrow$ refuse or EXPLAIN, not silent dense cosine substitution.
- **HyDE-style rewrite (Gao et al.).** Logical Rewrite_{hyde} generates hypothetical document(s) via an instruction-following LM, embeds with the corpus contrastive encoder, and searches by doc–doc similarity [Gao et al., 2023] [ESTABLISHED] as mechanism (journal 0032; unreproduced AUTHOR tables [PROVISIONAL]). RQL surface REWRITE HYDE / CTE packaging and token-cost attributes remain [HYPOTHESIS]; optional multi-sample mean ($\pm$ raw query embedding) follows Eqs. (6)–(8).
- **MMR diversify (Carbonell & Goldstein).** Logical $\text{Diversify}_{mmr}(\lambda)$ greedily reorders an evidence list by maximal marginal relevance [Carbonell and Goldstein, 1998] [ESTABLISHED] as formula (journal 0033; unreproduced AUTHOR Table 1 / SUMMAC [PROVISIONAL]). RQL DIVERSIFY MMR packaging, default λ , and Sim choices remain [HYPOTHESIS]; place after Search/Fuse—do not confuse with Fuse_* .
- **Similarity join.** $\text{VSimJoin}_{\theta}(E_1, E_2)$ may compile to nested Search_{dense} / range-iterator joins when ITERATIVE (or equivalent) exists [Zhang et al., 2023] [HYPOTHESIS]; otherwise to blocked nested loops with explicit cost — not claimed implemented.

5.5 Semantics sketch

Channels remain separate until fuse. Mandatory predicates in $P_{acl} \subseteq P$ must be enforced by physical plans that cannot strip them client-side [HYPOTHESIS]. Rewrite operators carry explicit token/latency cost attributes for the optimiser. Optional plan-level budget attributes (RECALL TARGET / LATENCY) are [HYPOTHESIS] packaging inspired by BlinkDB’s dual ERROR/TIME contracts and ELP selection [Agarwal et al., 2013] [ESTABLISHED] as AQP mechanism (journal 0031)—mapped here to retrieval effort knobs, not to stratified SQL samples (§6.4).

5.6 Delineation from survey material

RRF [Cormack et al., 2009] (formula + $k=60$; journal 0011), Condorcet-fuse [Montague and Aslam, 2002] (journal 0018), Bruch CC/TM2C2 linear fusion [Bruch et al., 2023] (journal 0016), Chen zero-shot RRF-vs-interp [Chen et al., 2022] (journal 0017), ColBERT MaxSim/late interaction [Khattab and Zaharia, 2020] (journal 0012), PLAID centroid interaction/pruning [Santhanam et al., 2022] (journal 0014), MUVERA FDE reduction [Dhulipala et al., 2024] (journal 0013), HyDE hyp-doc rewrite [Gao et al., 2023] (journal 0032), MMR diversify [Carbonell and Goldstein, 1998] (journal 0033), HNSW [Malkov and Yashunin, 2018], ACORN’s predicate-subgraph idea [Patel et al., 2024], VBASE’s relaxed-monotonicity iterators [Zhang et al., 2023], and Filtered-DiskANN’s label-aware graphs [Gollapudi et al., 2023] are prior art (multimodal notes: journals 0006, 0009–0014, 0016–0018, 0031–0033). **Original to this draft:** packaging them as a single portable retrieval IR with explicit physical filter modes and budget contracts aimed at RAG reliability [HYPOTHESIS].

5.7 Draft LogicalPlan schema freeze (Hypothesis IR)

A machine-checkable JSON Schema (draft 2020-12) for LogicalPlan is frozen at package path `schemas/logical-plan.schema.json` (version 0.1.0-draft; journal 0021). It encodes the kernel operators above plus optional capability annotations aligned to the Sep 2026 vendor matrix (§7.3). This freeze is [HYPOTHESIS] IR packaging—not a published standard—and does not promote unreproduced AUTHOR metrics. Validated examples live under `schemas/examples/`; the harness `experiments/harness/validate_plans.py` checks them with `jsonschema`.

5.8 Toy textual RQL frontend (Hypothesis prototype)

A minimal parser under `experiments/harness/rql_parser/` (journal 0022) maps a *tiny* textual subset—RETRIEVE/SEARCH {DENSE,BM25,LATE/COLBERT}, opaque WHERE predicates, optional ACL_HARD, FUSE RRF/LINEAR, K/CANDIDATES/LIMIT—onto LogicalPlan JSON conforming to the draft schema. It is an existence proof that a textual frontend can target the frozen IR; it is **not** a full SQL engine and rejects `EMBED/WITH/RERANK/...` with explicit errors [HYPOTHESIS].

6 Physical planning (original sketch)

This section is [HYPOTHESIS] unless a citation marks established prior art.

6.1 Plain language first

The logical plan says *what* to retrieve; the physical plan chooses *how* on a given backend:

- **Filter placement:** apply predicates before ANN (PRE), after ANN (POST), during ANN iteration (ITERATIVE), on a predicate subgraph (SUBGRAPH), via a specialised label graph (SPECIALIZED), via partitions (PARTITION), or via a multi-algorithm router (ROUTER/AUTO).
- **Fusion placement:** native RRF/weighted fusion when the vendor advertises it; otherwise an explicit client shim (never a silent ACL bypass).
- **Late-interaction ladder:** native MaxSim → PLAID-class prune → MUVERA FDE → MIPS+MaxSim rerank; else fail closed.
- **Budgets:** optional recall/latency targets steer effort knobs (`ef/nprobe/candidates`), inspired by BlinkDB’s dual contracts [Agarwal et al., 2013] [ESTABLISHED] as AQP doctrine—not transferred ϵ intervals.

Adapters must *advertise* capabilities; missing modes force documented fallbacks [HYPOTHESIS].

6.2 Physical filter modes

Grounded in the FANNS survey taxonomy of Lin et al. [Lin et al., 2025] (pruning strategies VSP/VJP/SJP/SSP; families A1–A17; journal 0019) and multimodal re-reads of ACORN [Patel et al., 2024], Filtered-DiskANN [Gollapudi et al., 2023], and VBASE [Zhang et al., 2023]:

$\text{FilterExec} \in \{\text{PRE}, \text{POST}, \text{ITERATIVE}, \text{SUBGRAPH}, \text{SPECIALIZED}, \text{PARTITION}, \text{ROUTER}, \text{AUTO}\}.$

Survey taxonomy $\rightarrow$ modes (packaging). Lin et al. argue that coarse pre-/post-/in-filtering is too coarse and propose a pruning-focused classification [ESTABLISHED] as *survey structure* (Figs. 1–2):

- $\text{PRE} \leftarrow \text{SSP}$ / Pre-Filtering family (A12): exact scalar filter then scan the survivor set [Lin et al., 2025].
- $\text{POST} \leftarrow \text{VSP}$ / Post-Filtering family (A1): ANN top- K' then predicate; K' estimation is brittle under unknown selectivity [Lin et al., 2025, Patel et al., 2024].
- $\text{ITERATIVE} \leftarrow \text{VBase}$ (A2) as a VSP refinement with filter-aware updates and relaxed-monotonicity stopping [Lin et al., 2025, Zhang et al., 2023]; Volcano Open/Next/Close when the backend advertises iterators.
- $\text{SUBGRAPH} \leftarrow \text{VJP}$ / ACORN-class predicate-subgraph traversal (A4; AIRSHIP A3 related) [Lin et al., 2025, Patel et al., 2024].
- $\text{SPECIALIZED} \leftarrow \text{VJP}$ equality/range graph families (Filtered-DiskANN A9; SeRF/iRangeGraph A10–A11 class) when the predicate matches advertised capabilities [Lin et al., 2025, Gollapudi et al., 2023].
- $\text{PARTITION} \leftarrow \text{SJP}$ workload/filter partitions with per-subset FANNS (Milvus-Partition A13; HQI A14; related multi-subset SJP) [Lin et al., 2025].
- $\text{ROUTER} \leftarrow$ system-level *combination of multiple FANNS algorithms* with per-query selection (survey open direction §6.3; selectivity cost-model switches in ADBV/Milvus/VBase; ACORN $\rightarrow$ PRE when highly selective) [Lin et al., 2025] [ESTABLISHED] as a *literature direction*; RQL opcode packaging [HYPOTHESIS].
- $\text{AUTO} \leftarrow$ choose among *advertised* modes using stats (selectivity, specificity, query–predicate correlation / ID–POD–OOD distribution factor [Lin et al., 2025], empty-result risk); **never** violate ACL hard constraints [HYPOTHESIS].

Label discipline: VSP/VJP/SJP/SSP and A1–A17 are [ESTABLISHED] survey vocabulary; FilterExec names and the planner policy that binds them are [HYPOTHESIS]. Author recall curves (e.g. survey Fig. 3) are **not reproduced here**.

Capability negotiation. Adapters must advertise which modes exist. Missing SUBGRAPH/SPECIALIZED/PARTITION forces a documented fallback — the algebra must not pretend universal support [HYPOTHESIS]. EXPLAIN should surface FilterExec, survey pruning_strategy when known, selectivity estimates, and optional distribution factor.

6.3 Cost vector

Unlike classical selectivity-only costing, plans expose a vector:

1. Recall proxy (candidate depth, filter empty-result risk, low-specificity risk [Gollapudi et al., 2023], distribution-mismatch risk [Lin et al., 2025]),

2. Latency (p50/p95 targets),
3. ACL safety (boolean hard constraint),
4. Token/dollar budget (embeddings, rerankers, rewrites).

6.4 Approximate budgets (BlinkDB-inspired)

BlinkDB [Agarwal et al., 2013] [ESTABLISHED] that approximate engines should expose *declarative* quality-or-latency contracts and select physical effort via an Error-Latency Profile (ELP): pilot small samples, project error ($\sim 1/\sqrt{n}$ for Table 2 aggregates) and latency ($\sim$ linear in rows read), then pick a multi-resolution sample (journal 0031). We steal that *contract + profile* doctrine for retrieval plans [HYPOTHESIS]:

- Surface options `OPTION RECALL TARGET / LATENCY` (and `PhysicalPlan` `budgets.recallTarget/latency`) declare soft constraints.
- The planner may raise/lower ANN effort (`ef/nprobe/candidates`), `FilterExec` mode, or rewrite/rerank depth to chase those constraints, surfacing choices in `EXPLAIN`.
- **Delineation:** BlinkDB ϵ is sampling error on SQL aggregates with closed-form variance; RQL “recall target” is an ANN/IR *recall proxy*—not a transferred confidence interval and not a measured claim in this draft.

6.5 Fusion and rewrite rules

Examples [HYPOTHESIS]:

- Independent channel searches $\rightarrow$ parallel schedule; choose fuse family per §6.6 [HYPOTHESIS] (policy grounded in Cormack/Montague/Chen/Bruch mechanisms [Cormack et al., 2009, Montague and Aslam, 2002, Chen et al., 2022, Bruch et al., 2023]).
- `Searchlate` compile ladder: native `MaxSim` multi-vector [Khattab and Zaharia, 2020] [ESTABLISHED] scoring; else `PLAID`-class centroid interaction + multi-stage prune [Santhanam et al., 2022] [ESTABLISHED] as mechanism (journal 0014) if `late_plaid` advertised; else `MUVERA FDE $\rightarrow$ MIPS + MAXSIM_RERANK` [Dhulipala et al., 2024] if `fde_mips` capability exists (journal 0013); else fail closed with `EXPLAIN` (do not silently substitute single-vector cosine).
- Low selectivity + `POST` $\rightarrow$ rewrite toward `ITERATIVE/SUBGRAPH/SPECIALIZED/PARTITION/over-fetch` depending on capabilities and predicate shape [Lin et al., 2025, Patel et al., 2024].
- High selectivity + cheap survivors $\rightarrow$ prefer `PRE` (SSP) or `ROUTER` policies that switch to `Pre-Filtering` [Lin et al., 2025].
- `VSimJoin` $\rightarrow$ prefer iterator nested loops when `ITERATIVE` exists [Zhang et al., 2023].

6.6 When `Fuserrf` vs `Fuselinear` (planner preference)

Chen et al. [Chen et al., 2022] and Bruch et al. [Bruch et al., 2023] disagree on which fuse wins empirically; the disagreement is [ESTABLISHED] as a literature conflict of *setups* (journal 0017 cite-chase): `Recall@1K` zero-shot `RRF` vs `NDCG TM2C2`, different linear forms, and different label budgets. We therefore treat the following as a planner [HYPOTHESIS] (not a reproduced metric claim):

- Prefer `Fuserrf` ($k=60$) when channel scores are incomparable/uncalibrated, or when no (or vanishing) labeled queries exist to tune α/ϕ — Chen-style zero-shot portability [Cormack et al., 2009, Chen et al., 2022] [ESTABLISHED] as mechanism.

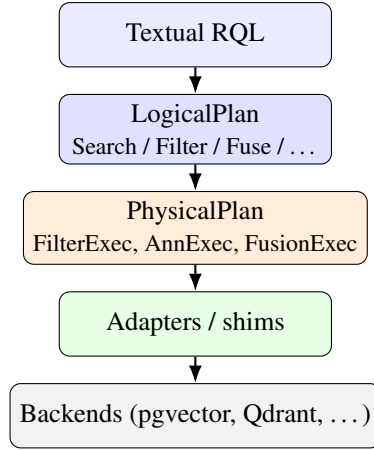


Figure 2: Hypothesized RQL compilation stack: text $\rightarrow$ logical plan $\rightarrow$ physical plan $\rightarrow$ adapters.

- Offer $\text{Fuse}_{\text{condorcet}}$ (pairwise-majority sort) as a rank-only Fuse-family sibling when majoritarian semantics are desired [Montague and Aslam, 2002] [ESTABLISHED] as mechanism (journal 0018); do not displace RRF as the portable default (Cormack AUTHORS prefer RRF on their suites) [HYPOTHESIS].
- Prefer $\text{Fuse}_{\text{linear}}(\alpha)$ with documented ϕ (TM2C2-class theoretical min–max when available) when scores are comparable and a small labeled set can tune α — Bruch-style [Bruch et al., 2023] [ESTABLISHED] as mechanism.
- Escalate to Fuse_{ltr} when query features or multi-channel weights justify learning beyond a single α [HYPOTHESIS].
- EXPLAIN must surface fuse family, k or α/ϕ or Condorcet comparator, and input channels.

6.7 Prototype physical planner stub (Hypothesis)

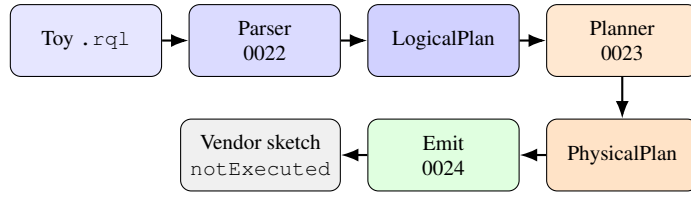
A deterministic stub under `experiments/harness/rql_planner/` (journal 0023) maps LogicalPlan JSON to PhysicalPlan JSON against docs-derived capability profiles (Qdrant / Elasticsearch / pgvector from the Sep 2026 vendor matrix; **not** live probes). Rules (Hypothesis packaging only): $\text{Filter} \rightarrow \text{FilterExec}$ mode via the toy selectivity/ACL chooser aligned with §6.2; $\text{Fuse}_{\text{rrf}} \rightarrow$ native FusionExec or explicit `ShimCast client_rrf`; $\text{Search}_{\text{late}} \rightarrow \text{LateInteractExec}$ (colbert default; optional PLAID/MUVERA rewrite flags when advertised). No fabricated latency or recall numbers; schema validation of planned outputs is a structural check only.

7 Compilation and adapters

[HYPOTHESIS] doctrine; *Substrait/polystore inspiration is [PROVISIONAL]/[ESTABLISHED] as prior systems. Vendor API cells below are [ESTABLISHED] from official docs (2026-09-16 survey); the claim that RQL compiles cleanly through them is [HYPOTHESIS].*

7.1 Layers

1. Parse textual RQL to LogicalPlan.
2. Optimise to PhysicalPlan with capability negotiation.
3. Emit backend calls through adapters (pgvector SQL, Qdrant Query API, Elasticsearch/OpenSearch DSL, Weaviate GraphQL, Milvus `hybrid_search`, ...).



*Profiles: qdrant / elasticsearch / pgvector (docs-derived).
CLI: experiments/harness/rql_pipeline.py — offline E2E (journal 0026).*

Figure 3: Offline end-to-end prototype pipeline (journals 0022–0024, 0026). All artifacts are files on disk; no live vector-database I/O. Hypothesis packaging; schema validation is a structural check only.

A prototype toy parser (journal 0022; §5.8) implements layer 1 for a schema-aligned subset only; the journal 0023 planner stub (§7.7) implements a rule-based layer 2; the journal 0024 emit stub (§7.8) sketches layer 3 request bodies; an offline end-to-end CLI (journal 0026; §7.9) glues layers 1–3 over toy examples—Hypothesis packaging, not a production compiler or live smoke.

7.2 Hello RQL walkthrough (implementer path)

Paths are relative to the repository root `rql-rag-query-language/`.

1. **Input:** `examples/toy/01-hybrid-rrf.rql` (Figure 1).
2. **Parse:** `experiments/harness/parse_rql.py parse examples/toy/01-hybrid-rrf.rql -validate` → LogicalPlan JSON conforming to `schemas/logical-plan.schema.json`.
3. **Plan:** `experiments/harness/plan_rql.py -profile qdrant ...` → PhysicalPlan JSON (FusionExec with native RRF on the Qdrant profile).
4. **Emit:** `experiments/harness/emit_rql.py ...` → Qdrant Query API-shaped JSON sketch labelled `notExecuted`.
5. **One-shot E2E:** `experiments/harness/rql_pipeline.py over examples/toy/*.rql × {qdrant,elasticsearch,pgvector} = 12/12 validated` (`experiments/results/e2e/run_validate`).

Unit tests live beside the harness (`test_rql_parser.py`, `test_rql_planner.py`, `test_rql_adapters.py`, `test_rrf_fusion.py`, ...). None of these steps requires a running vector database [HYPOTHESIS].

7.3 Established adapter surface survey (docs-only, Sep 2026)

A docs-only survey of public vendor APIs (journal 0020; companion package `docs/09-vendor-api-matrix.md`; OKF `knowledge/reads/vendor-api-matrix-2026-09/`) records, without live cluster calls:

- **Filter DSLs** differ (Qdrant JSON `Filter`; ES/OS Query DSL; Weaviate `where`; Milvus `expr`; pgvector SQL `WHERE`).
- **Filter+ANN composition** is not uniform: Weaviate documents pre-filtering; Elasticsearch documents `knn.filter` during approximate kNN; pgvector documents post-filter on approx indexes plus iterative scans ($\geq 0.8.0$); Milvus API text states `expr` before ANN; Qdrant emphasises leaf filter propagation; Redis documents hybrid `BATCHES/ADHOC_BF` modes.
- **RRF** is first-class in Qdrant (`rrf`), Elasticsearch (`retriever.rrf`), OpenSearch (`score-ranker RRF`), and Milvus (`RRFRanker`); Weaviate exposes `rankedFusion`; pgvector/Pinecone often require client/SQL shims.

- **Weighted / score fusion** appears as `MilvusWeightedRanker`, `Weaviate relativeScoreFusion+alpha`, `Qdrant dbsf/formula`, `ES boosts/weighted RRF`—semantics are *not* interchangeable.
- **Multi-vector / late interaction** is documented for Qdrant multivector rescore and Weaviate multi-vector (ColBERT-style); elsewhere often **UNKNOWN** or nested dense only.
- **EXPLAIN / plan visibility**: PostgreSQL `EXPLAIN` (pgvector) and ES Profile / OpenSearch hybrid score explanation are clearly documented; several others remain **UNKNOWN**, so RQL `EXPLAIN` will often be adapter-synthesised.

URLs and UNKNOWN cells are cited in the package matrix; this section does not invent undocumented APIs [ESTABLISHED] as a *surface survey*.

7.4 Hypothesis: RQL compiles to these shims

We [HYPOTHESIS] that a single RQL LogicalPlan can target the above surfaces via capability negotiation: map `Filter` $\circ$ `Searchdense` to the backend's PRE / POST+iterative / leaf-propagated / hybrid-batch mode; map `Fuserrf` to native RRF where present else an explicit client shim (ACL-marked); map `Fuselinear` only when score-normalisation contracts match; map `Searchlate` to native multivector paths or to the ColBERT→PLAID→MUVERA rewrite ladder. Live smoke verification remains a human P1 task—not claimed done here.

7.5 Capability negotiation

Profiles declare required features (hybrid+RRF, graph hop, late interaction, filter mode). Missing capabilities yield either a safe rewrite, a shim CAST (e.g. client-side fuse—explicitly marked expensive/unsafe for ACL), or a hard error [HYPOTHESIS].

7.6 Draft PhysicalPlan schema freeze (Hypothesis IR)

Companion JSON Schema `schemas/physical-plan.schema.json (0.1.0-draft; journal 0021)` freezes physical nodes used by capability negotiation: `FilterExec` modes {PRE,POST,ITERATIVE,SUBGRAPH,AnnExec,FusionExec,LateInteractExec(colbert/plaid/muvera),ShimCast, and optional recall/latency/ACL budgets. Capability flags on plans mirror the docs-only vendor matrix (§7.3) so missing RRF or late-interaction ads compile to explicit shims or fail closed—never silent dense cosine for `Searchlate` [HYPOTHESIS]. Schema freeze is [HYPOTHESIS] IR; vendor cells remain [ESTABLISHED] as a surface survey only.

7.7 Prototype Logical→Physical planner stub

Journal 0023 implements layer 2 of the compile stack as a [HYPOTHESIS] rule engine: capability profiles under `rql_planner/profiles/` encode docs-only flags (`rrfNative`, `annIterator`, `multiVectorLate`, ...) from §7.3; the planner emits schema-valid PhysicalPlan trees for the four toy-parsed logical examples across three profiles (12/12 validated). This is capability negotiation scaffolding—not a Cascades cost model; adapter emission is a separate sketch stub (§7.8).

7.8 Prototype adapter emit stub (Hypothesis sketches)

Journal 0024 implements layer 3 of the compile stack as a [HYPOTHESIS] emitter under `experiments/harness/rql`. PhysicalPlan JSON plus a docs-derived profile yields *vendor request sketches* only—Qdrant Query API-shaped JSON (`filter+prefetch/fusion=rrf` or `nearest`), Elasticsearch `retriever.rrf`

/knn(+filter) JSON, and pgvector SQL (ORDER BY embedding <=> ... LIMIT, with iterative-scan comments when FilterExec mode is ITERATIVE). Every artifact is labeled Hypothesis / approximate / notExecuted; the stub never opens sockets or hits live vector DBs. A smoke subset (hybrid-RRF + filtered-dense $\times$ three profiles = 6/6) is written under experiments/results/rql_adapters/. Live adapter smoke remains a human P1 task (protocol 03)—not claimed done here.

7.9 Offline end-to-end pipeline CLI

Journal 0026 glues layers 1–3 into a single offline command (experiments/harness/rql_pipeline.py): `toy .rql  $\rightarrow$  LogicalPlan  $\rightarrow$  plan(profile)  $\rightarrow$  PhysicalPlan  $\rightarrow$  vendor emit sketches`, writing a run folder under experiments/results/e2e/. Profiles qdrant/elasticsearch/pgvector; optional JSON Schema validation for both plan kinds; every artifact remains notExecuted (no live vector-DB I/O). Smoke on examples/toy/*.rql: four queries $\times$ three profiles = 12/12 schema-validated (experiments/results/e2e/run_validate.txt). This establishes that an offline E2E prototype exists as files-on-disk; it does **not** replace human P0 FANNS/dataset work or P1 live adapter smoke [HYPOTHESIS].

7.10 Substrait / Calcite IR adjacency

Source facts below are [ESTABLISHED] from Substrait docs (accessed 2026-09-17) and Begoli et al. [Begoli et al., 2018]; Cascades [Graefe, 1995] is cited for optimizer vocabulary. RQL packaging remains [HYPOTHESIS] adjacency—**not** a Substrait implementation or Calcite embedding (journal 0025).

Established prior systems. Substrait [sub, 2024] is an open, cross-language specification for *serialized relational/compute plans* (binary protobuf and text), complementary to SQL: producers and consumers exchange plans rather than SQL text. Plans are trees of relations (DAGs via reference relations) with emit/ordering, optional hints and constraints, and distribution/orderedness properties. The specification does **not** enforce a hard logical/physical split—operators such as hash join, exchange, and top- N are conventionally physical; the consuming system decides what counts as a physical plan. Extension points include YAML simple extensions (types and functions via extension URNs), AdvancedExtension (ignorable *optimization* hints vs semantic *enhancements*), and custom ExtensionLeafRel/ExtensionSingleRel/ExtensionMultiRel relations whose detail schemas are producer/consumer agreements.

Apache Calcite [Begoli et al., 2018] is an embeddable relational algebra and optimizer framework over heterogeneous sources: SQL parser or RelBuilder yields a tree of relational operators; pluggable rules, metadata providers, and planner engines rewrite that tree; adapters expose external schemas and push operators into backend *calling conventions*. Physical properties are modeled primarily as **traits** (ordering, partitioning, calling convention) rather than a mandatory dual logical/physical operator class hierarchy; an *enumerable* convention supplies iterator-based client operators when a backend cannot execute an op. Calcite’s cost-based engine uses Volcano-like dynamic programming over equivalence sets [Begoli et al., 2018]; Cascades [Graefe, 1995] supplies classic vocabulary (task-based search, memoization, property *enforcers*) behind many “Cascades/Volcano-style” citations.

Hypothesis: RQL is inspired-by, not identical. We [HYPOTHESIS] that RQL’s draft LogicalPlan/PhysicalPlan JSON (0.1.0-draft; journal 0021) and capability-negotiated compile stack (journals 0022–0024, 0026) are *Substrait-inspired* (portable plan IR; domain ops as future extension relations; adapters as plan consumers) and *Calcite-inspired* (algebra tree + physical strategies via capability/trait negotiation; pushdown vs explicit ShimCast fallback). Honest non-claims: RQL does **not** implement Substrait protobuf wire compatibility; is **not** a Calcite extension or VolcanoPlanner; and journal 0023 is **not** a Cascades cost model. A future retrieval-oriented Substrait dialect (custom relations for Search_{late} / Fuse_{rrf} / FANNS filter modes) remains optional future work—not claimed done here.

8 Applications to RAG patterns

Patterns below map to example programs in the companion repo (`examples/*.rql`).

- Dense + ACL metadata filter (reliability-first pushdown).
- **Hybrid BM25+dense with RRF / linear:** $\text{Fuse}_{rrf}(k=60)$ over channel-local rankings [Cormack et al., 2009] [ESTABLISHED] as the portable default when scores are incomparable; optional cross-encoder Rerank afterward (`examples/02-hybrid-rrf.rql`). Prefer $\text{Fuse}_{linear}(\alpha)$ (Bruch-style normalised convex combination [Bruch et al., 2023] [ESTABLISHED] as mechanism, journal 0016) when calibrated scores + small labels exist; keep RRF under zero-shot / uncalibrated scores [Chen et al., 2022] [ESTABLISHED] as the competing literature recommendation (journal 0017; preference rule [HYPOTHESIS], §6.6); optional $\text{Fuse}_{condorcet}$ (pairwise-majority sort) [Montague and Aslam, 2002] [ESTABLISHED] as mechanism (journal 0018) for majoritarian A/B without replacing RRF; escalate to Fuse_{ltr} for multi-feature learned fuse [HYPOTHESIS] (`examples/03-hybrid-linear-weights.rql`, `12-fuse-learned-explain.rql`).
- **HyDE rewrite as plan node:** Rewrite_{hyde} (hyp-doc generation $\rightarrow$ contrastive encode $\rightarrow$ dense doc-doc retrieve) [Gao et al., 2023] [ESTABLISHED] as mechanism (journal 0032); RQL REWRITE HYDE / CTE packaging + token cost under EXPLAIN [HYPOTHESIS] (`examples/10-hyde-rewrite-budget`) compose with BlinkDB-inspired budgets, journal 0031). Generic multi-query / DMQR siblings remain [PROVISIONAL].
- **MMR diversify as plan node:** $\text{Diversify}_{mmr}(\lambda)$ (relevance–novelty greedy reorder) [Carbonell and Goldstein, 1998] [ESTABLISHED] as formula (journal 0033); RQL DIVERSIFY MMR packaging [HYPOTHESIS] (post-retrieve / post-fuse; not a fuse op).
- Parent expansion and bounded multi-hop graph retrieval [PROVISIONAL].
- **Late-interaction search:** logical Search_{late} with MaxSim-sum semantics [Khattab and Zaharia, 2020] [ESTABLISHED]; optional planner rewrite toward PLAID-class centroid interaction/pruning [Santhanam et al., 2022] [ESTABLISHED] as mechanism (journal 0014) or MUVERA FDE $\rightarrow$ MIPS+MaxSim rerank [Dhulipala et al., 2024] when the adapter lacks native multi-vector support [HYPOTHESIS] (`examples/09-colbert-late-interaction.rql`).

RQL’s value is making these patterns *comparable* under EXPLAIN and shared eval harnesses [HYPOTHESIS] — not inventing RRF or MaxSim themselves [ESTABLISHED].

9 Evaluation: what we measured vs what remains

This section separates feasibility evidence already on disk from citation-grade ANN/RAG work that is not claimed. No fabricated metrics.

9.1 What we measured (feasibility / plumbing)

- **Deterministic unit tests (not IR benchmarks):** RRF on fixed lists (`experiments/harness/test_rrf_fusion` $\rightarrow$ `experiments/results/rrf_unit_test.txt`); weighted linear/CC on fixed toy scores (`test_linear_fusion.py` $\rightarrow$ `experiments/results/linear_fusion/`); MaxSim toy check (`test_maxsim_late.py`); filter-strategy chooser (`test_filter_strategy_chooser.py`).
- **Schema + compile stack:** LogicalPlan/PhysicalPlan JSON Schema 0.1.0-draft validated by `validate_plans.py`; toy parser 4/4 (journal 0022); planner stub 4 logical $\times$ 3 profiles = 12/12 (journal 0023); adapter emit smoke hybrid-RRF + filtered-dense $\times$ 3 = 6/6 sketches (journal 0024); offline E2E CLI 4 queries $\times$ 3 profiles = 12/12 schema-validated (`experiments/results/e2e/run_validate` journal 0026).

- **Colab synthetic FANNS PRE/POST smoke** (journal 0029–0030): T4 GPU, faiss-gpu 1.15.1; $N=50,000$, $d=128$, seed 42; selectivities $\{0.01, 0.05, 0.1, 0.5\}$; artifacts under `experiments/results/fanns/c`
Honest label: plumbing + qualitative PRE vs POST latency/QPS *shape* on synthetic data only. **Not** P0; **not** SIFT1M; Recall@10=1.0 on this easy setup is **not** a citation-ready ANN quality claim.

Planned-run printer `print_planned_runs.py` emits `score=NOT_RUN` only—by design.

9.2 What remains for citation-grade ANN / RAG

- **P0 — large-set filtered ANN:** SIFT1M (or another licensed vector set) with ground-truth k NN; PRE/POST/ITERATIVE/(SUBGRAPH if available) sweeps; report recall@ k and latency with library versions in `ENV.txt`. Protocol: `experiments/protocols/01-fanns-microbench.md`; checklist: `experiments/PAUSE_CHECKLIST.md`.
- **P1 — live adapters:** docker/smoke against real Qdrant, Elasticsearch/OpenSearch, and pgvector using emitted sketches; ACL fail-closed checks. Protocol 03.
- **Datasets + judgments:** BEIR/MS MARCO-style collections under license; synthetic metadata predicates for selectivity sweeps; human or shared judgments for RAG end-to-end—none claimed here. Protocol 02; `experiments/HUMAN_TODO.md`.
- **Cost-model validation:** measured selectivity/latency models for AUTO FilterExec—not implemented.

9.3 Metrics (when P0/P1 exist)

Recall@ k , nDCG@ k , latency p50/p95, throughput (QPS) where applicable, **plan stability** (same RQL $\rightarrow$ same logical plan hash), and ACL violation count (must be zero). Separate **[AUTHOR-ONLY]** citations from **REPRODUCED** rows in any future result table.

9.4 Baselines (planned)

1. Hand-written SDK glue (LangChain/LlamaIndex-style scripts).
2. Vendor filter+hybrid APIs directly.
3. pgvector SQL hybrid patterns.
4. Fixed `FILTER_MODE` ablations vs AUTO (once implemented).

9.5 Ablation matrix (planned)

Channels $\in \{\text{dense, bm25, both}\}$; fuse $\in \{\text{RRF, linear}\}$; filter mode $\in \{\text{PRE, POST, AUTO}\}$; rerank on/off; rewrite on/off.

No end-to-end RAG, BEIR, or human-judgment metrics are claimed. No licensed large-set FANNS P0 is claimed.

10 Limitations and threats to validity

Sharpened for a technical-report audience: what can go wrong if a reader over-reads this draft.

- **Prototype $\neq$ production compiler.** The offline parse/plan/emit/E2E path is a Hypothesis existence proof (12/12 toy schema validations). It is not Cascades cost-based optimisation, not Substrait wire-compatible, and not load-tested.

- **Literature coverage.** Multimodal Pass 1–5 notes exist for ACORN, VBASE, Filtered-DiskANN, FANNS survey, RRF, Condorcet, Bruch, Chen, ColBERT, PLAID, MUVERA, HyDE, MMR, BlinkDB, and Substrait/Calcite adjacency (journals 0006, 0009–0014, 0016–0019, 0025, 0031–0033). Other citations (Indri, BigDAWG, some vendor corners) remain **[PROVISIONAL]**. Author empiric tables throughout remain **[AUTHOR-ONLY]** / unreproduced.
- **ANN nondeterminism.** Recall is implementation- and parameter-dependent; a portable IR cannot claim bit-identical results across HNSW libraries or vendor builds **[ESTABLISHED]** as a systems fact.
- **Adapter impedance.** Not all backends expose subgraph search, native RRF, late interaction, or EXPLAIN. Client shims may hide latency and—if mis-marked—ACL risk; capability negotiation must fail closed **[HYPOTHESIS]**.
- **Docs-only vendor matrix.** Journal 0020 cells are from public documentation (Sep 2026), not live cluster probes; APIs drift.
- **Synthetic FANNS is not P0.** Colab $N=50k$ Gaussian smoke must not be cited as filtered-ANN quality or as an ACORN/FANNS table reproduction.
- **Optimism bias.** Algebra elegance and schema-valid plans do not guarantee planner quality without measured cost models and workloads.
- **External validity for RAG.** No generator-in-the-loop evaluation, no faithfulness/citation metrics, and no production A/B evidence are claimed.

11 Conclusion and next steps

We framed RQL as a **portable retrieval-plan IR** for RAG—algebra plus physical planning under recall/latency/ACL—rather than as cosmetic SQL for vectors **[HYPOTHESIS]**. Established priors ground the pieces: FANNS filter strategies [Lin et al., 2025, Patel et al., 2024, Zhang et al., 2023, Gollapudi et al., 2023]; Fuse-family mechanisms [Cormack et al., 2009, Montague and Aslam, 2002, Bruch et al., 2023, Chen et al., 2022]; late-interaction ladders [Khattab and Zaharia, 2020, Santhanam et al., 2022, Dhulipala et al., 2024]; HyDE rewrite [Gao et al., 2023]; MMR diversify [Carbonell and Goldstein, 1998]; BlinkDB-style contracts [Agarwal et al., 2013]; Substrait/Calcite adjacency [sub, 2024, Begoli et al., 2018]. **Original packaging** is the single IR, capability-negotiated compile stack, and honesty discipline—not the underlying IR formulas themselves.

Feasibility evidence on disk: draft schemas 0.1.0-draft, docs-only vendor matrix, toy parser/planner/emit, and offline E2E 12/12 schema-validated runs. Colab synthetic FANNS remains plumbing only.

11.1 Actionable next steps for implementers

1. **Run the toy stack** (Section 7.2): `rql_pipeline.py` on `examples/toy/*.rql`; read emitted sketches under `experiments/results/e2e/`.
2. **Extend the grammar** only with schema-first changes: update `schemas/*.schema.json`, then parser/planner/emit tests—keep rejecting unsupported ops explicitly.
3. **P0:** land a licensed SIFT1M (or equivalent) FANNS microbench with ground truth; store `metrics.json` + `ENV.txt` + plans beside scores (`experiments/PAUSE_CHECKLIST.md`).
4. **P1:** live-adapter smoke from emit sketches against dockerised Qdrant/ES/pgvector; assert ACL fail-closed.

5. **Optional research:** cost-model stats for AUTO; Substrait extension relations for Search_{late} / Fuse_{errf} / FilterExec—inspired-by, not claimed done.
6. **Do not** cite synthetic Colab Recall@10 or AUTHOR paper tables as this project’s results.

This draft aims to be *arXiv-ready as a proposal + prototype technical report* once P0 artifacts and a short final edit pass land; it is **not** yet a venue-complete systems or IR evaluation paper.

A Reproducibility checklist

Companion to Section 9: artifact layout for measured plumbing and for future P0/P1 runs.

1. Scripts and configs under `experiments/` with pinned dependency files when benchmarks are added.
2. Dataset license notes and download digests (**HUMAN_TODO**: `experiments/protocols/02-datasets.md`).
3. Seeded RNGs where stochasticity exists; report ANN library versions in `ENV.txt`.
4. Store logical/physical plan JSON beside metrics (no orphaned scores).
5. Separate [**AUTHOR-ONLY**] citations from REPRODUCED runs in result tables.
6. Research notes: journal + OKF-shaped bundles for paper reads.
7. Hand-off list for compute/credentials beyond the agent sandbox: `experiments/HUMAN_TODOS.md`.
8. Offline E2E validation log: `experiments/results/e2e/run_validate.txt` (12/12 toy; notExecuted).

A.1 Colab synthetic FANNS smoke (in-repo, not P0)

Run id `colab_synth_20260916_233628` (Europe/Dublin wall clock aligned with journal 0029):

- Path: `experiments/results/fanns/colab_synth_20260916_233628/` (`metrics.json`, `ENV.txt`, `plans/`).
- Colab notebook URL registry: `experiments/colab/LINKS.md`.
- Backend: `faiss-gpu` 1.15.1 on T4; synthetic Gaussian vectors only.
- **What it supports:** harness wiring and a qualitative PRE $\ll$ POST latency/QPS shape on this synthetic setup.
- **What it does not support:** P0 claims, SIFT1M completion, or citation-ready filtered-ANN recall tables.

A.2 Companion reproducibility repository

Runnable code, dataset registry (no multi-GB binaries in git), synthetic fixtures, and small result smoke fixtures are maintained in a **separate** companion repository so this research tree can stay narrative-first (thesis, journal, OKF reads):

- URL: <https://github.com/vijaykumarjob0701/rql-repo>

- Research narrative / schemas-as-spec: <https://github.com/vijaykumarjob0701/rql-rag-query-lan>
- Evidence split: see root EVIDENCE.md in both repos (2026-09-17 Europe/Dublin audit).

P0 FANNS (licensed large sets + ground truth), live adapter smokes, and RAG judgments remain human/Colab hand-offs; the companion repo holds download scripts and pinned offline package wiring only.

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